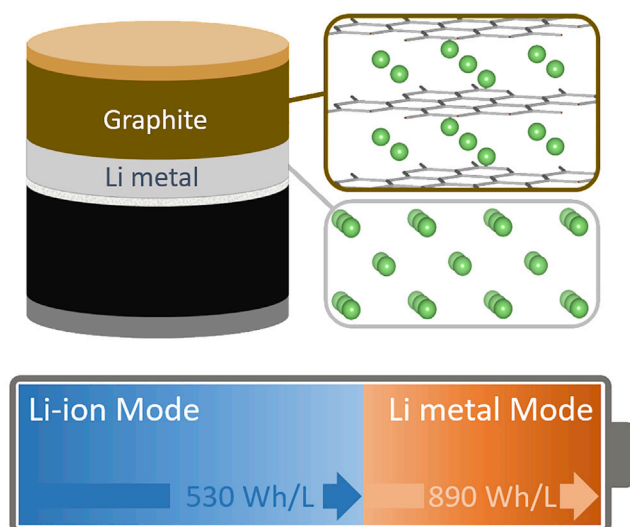


Article

Cycling Lithium Metal on Graphite to Form Hybrid Lithium-Ion/Lithium Metal Cells

Li-ion/Li Metal Hybrid Cell



Purposely plating lithium metal on a graphite anode in an optimized electrolyte enables hybrid lithium-ion/lithium metal cells that deliver 20% increased energy density compared with conventional lithium-ion cells. Hybrid cells can be operated highly reversibly in lithium-ion mode for low capacity utilization and charged fully to harness the extra capacity of the lithium metal for extended operation.

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HIGHLIGHTS

Cycling lithium metal on graphite enables hybrid lithium-ion/lithium metal cells

Hybrid cells deliver 20% higher energy density than conventional lithium-ion cells

Optimized electrolyte and pressure enable reversible lithium plating on graphite

Cells can operate in lithium-ion mode with periodic lithium metal cycles

Article

Cycling Lithium Metal on Graphite to Form Hybrid Lithium-Ion/Lithium Metal Cells

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SUMMARY

A hybrid anode cell design is proposed involving lithium metal plating on top of graphite that provides a 20% increase in energy density over conventional lithium-ion cells. Pouch cells with hybrid graphite-lithium metal anodes cycled with conventional electrolytes fell below 80% capacity in under 15 cycles. However, with a dual-salt electrolyte and applied mechanical pressure optimized for lithium metal cycling, hybrid cells achieved over 150 full (100% utilization) cycles before falling below 80% capacity with a CE of 99.6% for lithium metal plating on graphite. We also found that intermittent high energy (100% utilization) cycles utilizing lithium metal can be dispersed among hundreds of conventional lithium-ion cycles where only the graphite is utilized. Operating the cell with this intermittent protocol shows minimal impact to the underlying graphite capacity. Therefore, these hybrid cells can operate well in “lithium-ion mode” with periodic high energy full cycles accessing the lithium metal capacity.

INTRODUCTION

In an effort to increase the energy density of lithium-ion cells, many researchers are focused on developing new high-voltage or high-capacity electrode materials.^{1–3} On the cathode side, there is a continual push to increase the Ni content of $\text{Li}[\text{Ni}_{1-x-y}\text{Mn}_x\text{Co}_y]\text{O}_2$ (NMC) and $\text{Li}[\text{Ni}_{1-x-y}\text{Co}_x\text{Al}_y]\text{O}_2$ (NCA) layered oxide materials to increase capacity.^{4,5} For the anode, significant efforts have focused on high-capacity silicon alloy materials or lithium metal.^{2,6} In each case, the increased energy density offered by these new materials comes at a cost, mainly, decreased lifetime, decreased safety, or both. The lithium metal anode is a perfect illustration of this trade-off. Many believe that replacing graphite with lithium metal is the most promising approach to achieve major improvements in energy density over lithium-ion.⁷ However, unlike graphite, lithium metal shows huge volume expansion with cycling, which prevents the formation of a robust and stable solid electrolyte interphase (SEI) in cells with liquid electrolytes. An incomplete SEI that is constantly reforming leads to continual electrolyte consumption, lithium inventory loss, and the formation of dendritic lithium morphology prone to becoming electronically isolated.^{3,8,9} This poor morphology not only contributes to a significant reduction in cell lifetime but also presents safety concerns due to potential cell shorting.

The challenges of the lithium metal anode highlighted above can be obscured when evaluating laboratory lithium metal cells. A coin cell with large (10×) lithium metal excess and flooded electrolyte masks the issues of lithium inventory loss and electrolyte consumption leading to stable operation for many hundreds or even thousands

Context & Scale

Increasing the energy density of batteries will decrease the cost of electric vehicles and enable longer driving ranges. Replacing the graphite anode in conventional lithium-ion cells with lithium metal significantly increases energy density. However, lithium metal anodes are plagued with rapid capacity loss and short cell lifetime. To develop a high energy density cell with longer lifetime, we propose a hybrid lithium-ion/lithium metal cell that is achieved by purposefully plating lithium metal on graphite. Although unwanted lithium plating is normally a degradation mechanism in conventional lithium-ion cells, we achieve reversible lithium plating on graphite with an optimized dual-salt electrolyte. Furthermore, since batteries are often not cycled to 100% of their capacity, these hybrid cells can be operated in lithium-ion mode with little degradation for the majority of their life with periodic fully charged lithium metal cycles for extended capacity.

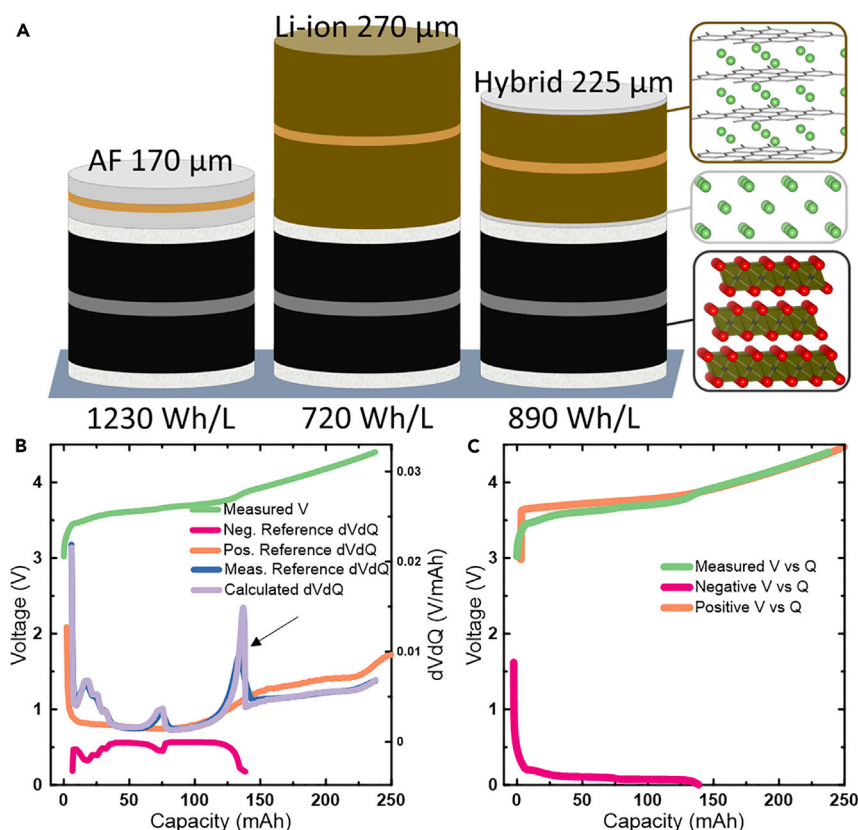


Figure 1. Hybrid Lithium-Ion/Lithium Metal Cells

(A) Schematic illustrating the thickness of the cell stack for anode-free lithium metal, conventional lithium-ion, and hybrid lithium-ion/lithium metal cells based on the electrode loadings from the pouch cells used in this study.

(B) Voltage and dV/dQ versus capacity for the first post formation charge to 4.4 V of a hybrid cell containing LDBF electrolyte.

(C) Voltage versus capacity plots demonstrating electrode capacity alignments after the first post-formation charge cycle to 4.4 V in LDBF electrolyte. (LDBF: 1 M LiDFOB 0.4M LiBF₄ FEC:DEC 1:2).

of cycles.^{10,11} However, the development of practical high energy density lithium metal cells requires the use of limited electrolyte (<3 g Ah⁻¹) and limited excess lithium (<1.5× excess).^{12–14} Under these more challenging conditions the cycle life of lithium metal cells is reduced substantially.¹⁴ This is best illustrated with the anode-free configuration, which has zero excess lithium and the highest possible energy density for a lithium metal cell. Anode-free cells, even those with optimized electrolyte formulations, often fall below 80% capacity retention in 40 cycles or fewer.^{11,15–18}

This significant trade-off between energy density and cycle life is also common to high-capacity silicon anode materials.^{19,20} With the silicon anode, one way to address this issue is with the partial incorporation of small amounts of silicon (5 wt %, 10 wt %, etc.) into a conventional graphite anode. With small amounts of silicon in graphite, the increase in energy density is obviously less than the pure alloy anode; however, the overall electrode volume expansion is also much less. Furthermore, since the delithiation potential for silicon is higher than graphite, the silicon portion is only accessed on deep discharge cycles, so the cell lifetime is much improved.

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Table 1. Parameters of a Lithium-Ion, Hybrid, and Anode-Free Cell Stack Based on the Electrode Loadings Used in This Work

	Li-Ion	Hybrid	Anode Free
Stack mass (mg/cm ²)	64.31	55.02	43.82
Stack thickness (μm)	272.50	224.50	167.00
Average voltage	3.70	3.76	3.90
Stack volumetric energy density (Wh/L)	720.55	890.05	1,231.08
Normalized stack volumetric energy density	1.00	1.24	1.71
Stack specific energy density (Wh/kg)	305.32	363.17	469.17
Normalized stack specific energy density	1.00	1.19	1.54

Based on this silicon compromise, we wondered if the same approach could be applied to lithium metal, that is, a hybrid anode containing both lithium metal and graphite. The simplest way to achieve this would be to build a lithium-ion cell with a negative/positive (n/p) capacity ratio much less than 1 so the graphite anode reaches its full capacity at a voltage well below the normal upper cutoff voltage of a properly balanced cell. Charging beyond this point would result in additional capacity being plated as lithium metal on top of the graphite. The amount of lithium plated could be controlled by the original cell balance or by adjusting the voltage to which the cell is charged. A cell stack schematic of this hybrid format is shown in [Figure 1A](#) along with the cell stacks of a conventional lithium-ion and anode-free lithium metal cells for comparison. [Table 1](#) shows the stack information as well as the stack energy densities of each cell format, with a more detailed list of parameters shown in [Table S1](#). The hybrid cell offers an intermediate energy density, less than anode-free, but greater than lithium-ion. Due to the increased average voltage and decreased stack thickness and weight, the hybrid format boasts a volumetric energy density (per L) increase of 25% and a specific energy density (per kg) increase of 20% over conventional lithium-ion cells.

This new anode concept presents many questions, most importantly: can lithium metal be reversibly cycled on top of or within a graphite anode? Plating on graphite is often considered a common failure mechanism that kills lithium-ion cells.²¹ Additionally, will these hybrid cells perform better in electrolytes optimized for lithium-ion operation or those optimized for lithium metal cycling? In this report we attempt to answer these and other questions through the testing and analysis of hybrid lithium-ion/lithium metal pouch cells. The morphology of lithium metal plated on graphite will also be investigated along with the performance of these hybrid cells on various charging protocols to determine the feasibility of this cell design.

RESULTS

Lithium Metal/Graphite Hybrid Cell Design

Conventional lithium-ion pouch cells utilizing a single crystal $\text{Li}[\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}]\text{O}_2$ (NMC 532) cathode were selected for this work. However, these cells were designed to contain a thin graphite anode such that the n/p ratio would be well below one (here, $n/p = 0.6$). This low n/p ratio enables the hybrid anode as the graphite will be charged above its maximum capacity within the normal operating voltage of the cell causing lithium plating on the graphite surface. [Figure 1A](#) offers a comparison of stack thickness for cells with identical cathodes utilizing a zero excess lithium metal anode (anode-free, “AF”), a conventional graphite anode (Li-ion), and a hybrid graphite/lithium metal anode (hybrid). At the top of charge (4.4 V) the hybrid cell stack is ~20% thinner and ~15% lighter than the conventional lithium-ion design,

Table 2. Electrolyte Compositions Selected for Cycling in Hybrid Graphite/Lithium Metal Cells

Salt	Solvent	Additives	Abbreviation
1.5 M LiPF ₆	25:5:70 EC:EMC:DMC	2 wt % VC	2VC
1.5 M LiPF ₆	25:5:70 EC:EMC:DMC	2 wt % FEC + 1 wt % LFO ^a	2F1L
1 M LiDFOB ^b + 0.4 M LiBF ₄	1:2 FEC:DEC (v/v)	–	LDBF

^aLiPO₂F₂.

^bLiBF₂(C₂O₄).

as detailed in Table 1. Furthermore, since a portion of the capacity of the hybrid cell is achieved by plating lithium, the average voltage of the negative electrode during cycling will be lower and thus the full cell average voltage will be higher than for conventional lithium-ion cells. This increase in average voltage and decrease in stack thickness and mass leads to a volumetric and specific energy density improvement of 25% and 20%, respectively.

One of these hybrid cells, after formation to 4.4 V, was evaluated using differential voltage (dV/dQ) analysis to determine the voltage and capacity at which the graphite anode becomes fully lithiated, causing lithium metal to begin plating on the anode surface.^{22–24} Differential voltage analysis was introduced by Bloom et al. and Honkura et al.^{23,24} Our lab's software used to perform the dVdQ fitting and analysis has been previously described.²⁵ Figure 1B shows the measured voltage profile for the hybrid cell (green) along with its corresponding differential voltage curve (blue). Positive (orange) and negative (magenta) half-cell reference dVdQ curves are used to generate the calculated full cell curve (purple), which was fit to the measured data. Beyond the conventional features corresponding to graphite staging, the measured and calculated full cell dVdQ curves show a large peak at approximately 140 mAh (highlighted with black arrow). This corresponds to the capacity at which the potential of the negative electrode falls to 0.0 V indicating that the graphite is fully lithiated and subsequent capacity will cause lithium plating on the graphite surface. At this point there is also a noticeable inflection in the full cell voltage curve, which occurs at approximately 3.85 V. This information is summarized in the full and half-cell voltage curves shown in Figure 1C. For these hybrid cells the graphite has a capacity of 141 mAh and when the cell is charged to 4.4 V it delivers a capacity of 230 mAh. The difference between these two capacities (~90 or 1 mAh cm⁻²) is plated as lithium metal on the graphite surface.

Electrolytes Optimization for Hybrid Cell Cycling

Three different electrolytes, listed in Table 2, were selected to evaluate the cycling of the hybrid graphite/lithium metal cells. These formulations were chosen to evaluate if the hybrid cells performed best in electrolytes optimized for lithium-ion operation or those optimized for lithium metal cycling. The 2% vinylene carbonate (2VC) formulation was chosen as a conventional lithium-ion "control" electrolyte. 2% fluoroethylene carbonate + 1% LiBF₂C₂O₄ (2F1L) represents an optimized Li-ion formulation, which has been shown to exhibit excellent capacity retention in lithium-ion cells with the same single crystal NMC 532 cathode being utilized in this work.²⁶ Finally, 1 M lithium difluoro(oxalato)borate (LiDFOB) + 0.4 M lithium tetrafluoroborate (LiBF₄) (LDBF) is an electrolyte optimized for lithium metal cycling, which has shown improved capacity retention and lithium morphology in anode-free lithium metal cells.²⁷

Hybrid cells were cycled in each of the three electrolytes with varying upper cutoff voltages of 4.2, 4.3, and 4.4 V with the results shown in Figures 2A–2C. The cells

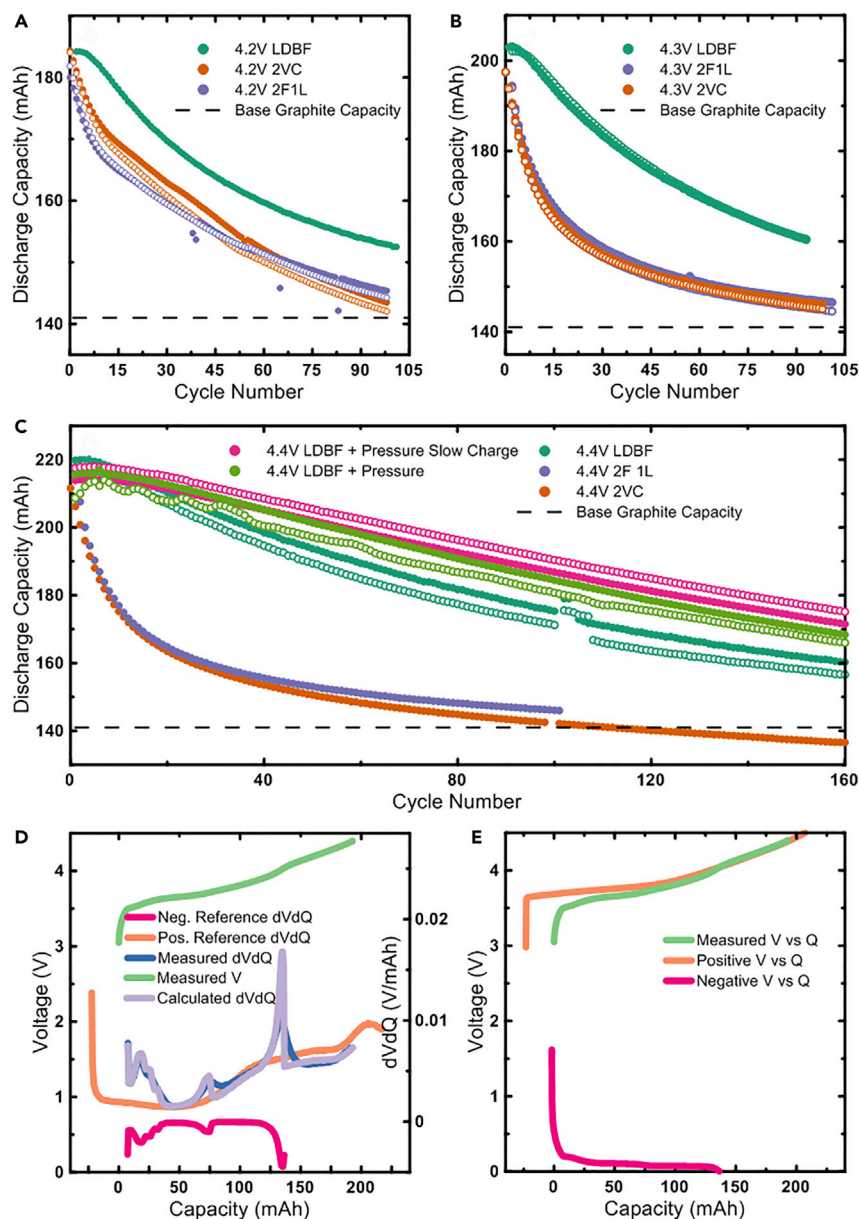


Figure 2. Performance of Lithium-Ion Electrolyte versus Lithium Metal Electrolyte

(A–C) Capacity retention of lithium-ion electrolytes 2VC and 2F1L and optimized lithium metal electrolyte LDBF in hybrid cells cycling between (A) 3.0–4.2, (B) 3.0–4.3, and (C) 3.0–4.4 V. Cells labeled “+pressure” were cycled with 1,200 kPa uniaxial pressure, all other cells were cycled under low-pressure (75 kPa). Cells labeled “slow charge” were cycled with a C/10 charge rate and C/2 discharge rate. All other cells were cycled with a C/5 charge rate and C/2 discharge rate. (D) Voltage and dV/dQ versus capacity for the first 50th charge to 4.4 V of a hybrid cell containing LDBF electrolyte. (E) Voltage versus capacity plots demonstrating electrode slippages after the 50th charge cycle to 4.4 V in LDBF electrolyte. Open and closed symbols of the same color represent pair cells. (LDBF: 1 M LiDFOB 0.4 M LiBF₄ FEC:DEC 1:2; 2F1L: 1.5 M LiPF₆ EC:EMC:DMC 25:5:70 2 wt % FEC 1 wt % LFO; 2VC: 1.5 M LiPF₆ EC:EMC:DMC 25:5:70 2 wt % VC)

were cycled at 40°C in a compliant plastic enclosure (Figure S1) where the pressure was maintained at approximately 75 kPa. It is clear from this cycling data that the lithium metal capacity plating on the graphite is sustained through cycling much

better in the optimized lithium metal LDBF electrolyte than either 2F1L or 2VC. This effect becomes more significant as we cycle to a higher upper cutoff voltage, which also increases the capacity of Li-metal cycled. Cycling smaller quantities of lithium is much less of a challenge so the 2VC and 2F1L can somewhat handle this scenario; however, with higher lithium metal capacities the performance gap between these two electrolytes and the LDBF becomes substantial.

The plating of lithium metal on graphite is a commonly cited failure mechanism for lithium-ion cells and so it would be reasonable to expect all of these hybrid cells to show rapid capacity fade, indicative of low Coulombic efficiency (CE) (Figure S2), and quickly lose their overbalanced lithium metal capacity. This is the case for the 2F1L and 2VC electrolytes, as these cells lost over 50% of their lithium metal capacity in less than 15 cycles. Conventional lithium-ion electrolytes with LiPF_6 salt, particularly without an abundance of fluoroethylene carbonate (FEC), do not adequately passivate lithium metal and generate a highly mossy lithium morphology.²⁸ As a result, capacity is quickly lost to forming dead lithium via mechanical isolation (lithium chunks breaking off the mossy surface) and to parasitic reactions forming the SEI. In stark contrast, the 4.4 V upper cutoff LDBF cells were able to retain over 50% of their lithium metal capacity after 80 cycles. The capacity loss here is relatively linear, indicating a constant rate of lithium loss that is not accelerated during cycling. LDBF electrolyte has been shown to enable pristine lithium morphology.²⁷ Therefore, failure is not accelerated by the formation of mossy lithium, which exacerbates the formation of dead lithium.

As this hybrid anode technology would require higher voltage cycling to show the maximum benefit in energy density by leveraging the most capacity from plated lithium, it is important that the cells could be cycled well at each voltage, and even better that the performance of cells with LDBF electrolyte increases with upper cutoff voltage in contrast to lithium-ion battery electrolytes. Figure S3 also shows the results of cycling hybrid cells to a 4.5 V upper cutoff in LDBF, which trended similarly to the 4.4 V cells. Interestingly, as can be seen in Figure S4, there was very little improvement in capacity retention in an electrolyte composed of 1.5 M LiPF_6 + 0.5 M LiDFOB + 0.3 M LiBF_4 in 25:5:70 ethylene carbonate:ethyl methyl carbonate:dimethyl carbonate (EC:EMC:DMC) when compared with 2F1L or 2VC. This suggested that LiDFOB and LiBF_4 as additives alone are not enough to achieve significant improvements in lithium metal capacity retention.

Aged hybrid cells that had undergone 50 cycles to 4.4 V in the LDBF electrolyte were again analyzed with differential voltage analysis to evaluate the changes since cycle 1 with the results shown in Figures 2D and 2E. There was no significant loss in the capacity of the graphite after cycling, as the capacity was still shown to be approximately 140 mAh, indicating that all the loss occurring was from the Li-metal capacity. The dV/dQ feature, indicating the onset of lithium metal plating, has shifted to approximately 4.05 V, which results from the slippage of the negative electrode, highlighted by the red arrow in Figure 2E. This graphite curve has slipped to the right by close to 30 mAh relative to the positive electrode. This confirms that the 30 mAh of capacity lost by the cell is due to inventory loss at the negative electrode. The full cell discharge condition is reached when the graphite is fully emptied, while the positive electrode cannot be filled to its maximum capacity due to the inventory loss. These results also demonstrate that the voltage at which plating begins will continually shift higher with cycling as more lithium metal capacity is lost and the graphite curve slips further to the right.

The effect of mechanical pressure on the cycling performance of the hybrid cells was also investigated. Applying pressure to lithium metal cells has been widely reported

Table 3. Average Cycling Efficiency for LDBF Cells under Different Conditions Calculated for Just the Lithium Metal Component as Well as the Full Cell Capacity

Cell	Charge and Discharge Rate	CE (Lithium Metal)	CE (Full Cell Capacity)
LDBF	C/5 C/2	99.18%	99.55%
LDBF + Pressure	C/5 C/2	99.49%	99.79%
LDBF + Pressure	C/10 C/2	99.58%	99.86%
LDBF Anode free + Pressure	C/5 C/2	99.54%	99.54%

to improve both lithium morphology and capacity retention.^{28,29} For the hybrid cells, an initial uniaxial pressure of 1,200 kPa was applied to the pouch cell during cycling via the pressure rig shown in [Figure S1](#). The capacity retention of hybrid cells under pressure with LDBF electrolyte is shown in [Figure 2C](#) along with that of the low pressure LDBF, 2F1L, and 2VC cells for comparison. Much like with pure lithium metal cells, capacity retention of the hybrid cells shows a substantial improvement with applied pressure as these cells achieved over 140 cycles while still maintaining 80% capacity retention and over 50% of the original lithium metal capacity. Hybrid cells were also tested under uniaxial pressure at 20°C after a “hot formation” procedure in which they were first cycled twice at 40°C before moving to 20°C testing ([Figure S5](#)). This was shown to significantly improve the 20°C performance of anode-free lithium metal cells.³⁰ Similarly, [Figure S5](#) shows that 20°C cycling of hybrid cells after hot formation can be sustained without increased degradation compared with cycling at 40°C.

Performance of the hybrid cells could be further improved by decreasing the charge rate. Lower charging current rates are generally considered beneficial for improving lithium metal morphology.³¹ Hybrid cells with the LDBF electrolyte and 1,200 kPa uniaxial pressure were cycled with a C/10 charge compared with the C/5 charge used for the previous cells, also shown in [Figure 2C](#). The slow charge cells showed a noticeable but modest benefit in capacity retention over the cells charged at C/5 maintaining 80% retention at 160 cycles.

The average CE over the first 100 cycles for each of the cell conditions tested with the LDBF electrolyte were calculated considering first only the estimated lithium metal capacity, and second the entire cell capacity, and is shown in [Table 3](#). The average CE for an anode-free lithium metal cell with the same cathode and same electrolyte formulation is included in [Table 3](#) for comparison. A comparison of the cycling performance between hybrid and anode-free lithium metal cells is shown in [Figure S6](#). An anode-free cell with LDBF electrolyte is also added for reference.

From the results of [Table 3](#), pressure is clearly a significant factor in the performance of the hybrid lithium metal cells. It is also interesting that similar lithium metal cycling efficiencies were achieved when plating on graphite in the hybrid cell and on copper in the anode-free cell. It should be noted that this comparison between the hybrid and anode-free cell is not perfect because the hybrid cells are cycling a smaller quantity of lithium, 1 mAh cm⁻² compared with 2.4 mAh cm⁻² in the anode-free configuration. Cycling a smaller quantity of lithium results in less volume expansion and generally favors higher CE values.

Surface Morphology of Hybrid Anodes

[Figure 3A](#) shows a hybrid graphite/lithium metal anode recovered from a LDBF pouch cell after the initial formation cycle to 4.4 V. In the anode regions, which

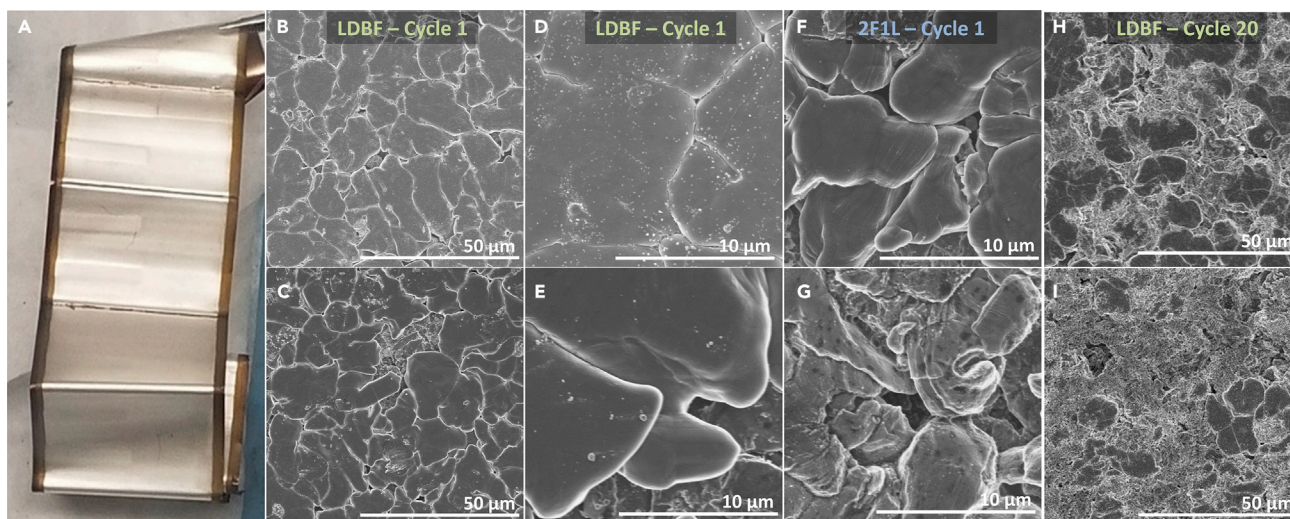


Figure 3. Morphology of Lithium Plated in Hybrid Cells

(A) Hybrid graphite/lithium metal anode removed from a pouch cell after the first C/20 formation cycle to 4.4 V in LDBF electrolyte.

(B–E) SEM images of hybrid graphite/lithium metal anodes in LDBF electrolyte after the first C/20 formation charge under low pressure (B and D) and high pressure (C and E).

(F and G) 2F1L hybrid anodes after the first C/20 formation charge under low pressure (F) and high pressure (G).

(H and I) LDBF hybrid anodes after the 20th C/5 charge cycle under low pressure (H) and high pressure (I). (LDBF: 1 M LiDFOB 0.4 M LiBF₄ FEC:DEC 1:2; 2F1L: 1.5 M LiPF₆ EC:EMC:DMC 25:5:70 2 wt % FEC 1 wt % LFO).

were directly opposite the cathode, a seemingly uniform lithium layer shows good coverage of the graphite underneath. In the anode “overhang” regions highly lithiated (yellow) graphite is observed. Lithium has diffused into this overhang region during the slow C/20 formation charge. The remaining images of Figure 3 show scanning electron microscopy (SEM) images taken of hybrid graphite/lithium metal anodes from cells charged to 4.4 V with and without applied pressure. Figures 3B–3G were taken after the initial C/20 formation cycle while Figures 3H and 3I were taken after 20 cycles (C/5 charge C/2 discharge).

In the LDBF electrolyte after the first formation cycle (Figures 3B–3E), the lithium shows a very flat and smooth morphology composed of large grains, which have packed together on top of the graphite. Coverage of the graphite is not complete as there are some areas where one can clearly see through the lithium grains to the graphite flakes below; the bottom right corner of Figure 3E is one example.

Nevertheless, the coverage is fairly uniform, which is somewhat surprising considering the lithium is plating on a porous graphite coating and not a flat current collector. These images after formation also clearly demonstrate that pressure has a positive impact on lithium morphology, causing grains to be plated in a more rounded, columnar manner. Figures 3B (high pressure) and 3C (low pressure) demonstrate the lithium in a cell with LDBF electrolyte forming grains fitting more closely together when pressure is applied, with less empty space showing the graphite surface underneath. Comparing Figure 3B with 3C, the average lithium grain size appears to be larger and less fragmented for the hybrid cells with pressure added. This agrees with previous studies indicating larger grain size results in superior continued morphology, dendrite suppression, and CE,³² which we see exhibited in the behavior of these cells. With pressure, these more rounded grains are generally of diameter 10–15 μm . With higher magnification (Figures 3D and 3E) the impact of pressure is more obvious with the lithium under the high-pressure condition showing

a smooth surface with interlocked grains, while the lithium under low pressure shows smooth grains that are less compact and not as tightly packed.

Figures 3F and 3G illustrate the morphology of hybrid anodes after the first formation cycling in the 2F1L electrolyte at high and low pressure, respectively. Under pressure, the 2F1L lithium grains are plating in a distinct layer on top of the graphite similar to the cell with LDBF; however, comparing Figures 3D and 3E, the grains of lithium are significantly smaller, with many pieces $<5\text{ }\mu\text{m}$ in diameter. The grains also do not fit together as compactly as those from the LDBF electrolyte. This agrees with the cycling data, as smaller average lithium grain size can result in larger anode surface area and increased parasitic losses leading to faster capacity loss.

Beyond the formation morphology it is also important to analyze cycled electrodes as large changes in lithium morphology can occur with cycling. Figures 3H and 3I show the SEM images at the top of charge after 20 cycles in LDBF electrolyte under high and low pressure, respectively. There is an obvious change in morphology after cycling, some flat lithium grains remain, but they appear smaller than those observed after formation. Additionally, now the flat lithium grains appear to be separated by more porous lithium that is difficult to distinguish from the graphite. In the low-pressure condition, the decline in the quality of lithium morphology is more severe as the regions of flat lithium grains are sparse. The deterioration of lithium morphology with cycling is common, however, it appears to be occurring at a more accelerated rate than a pure lithium metal anode under similar conditions.²⁷ The continued plating and stripping on a porous graphite layer may accelerate the change in morphology compared with plating and stripping on a flat metallic substrate in a lithium metal or anode-free cell. However, the CE of the hybrid cells summarized above, particularly those under high pressure, is very similar to the CE of an anode-free lithium metal cell under similar conditions. Beyond the difference in the amount of lithium being plated; this discrepancy between lithium morphology and CE behavior may be attributed to an added benefit of the graphite substrate. Archer et al. have demonstrated that lithium plated on a porous carbon cloth shows poor lithium morphology but relatively high CE.³³ They attribute this to the porous substrate “trapping” the plated lithium and preventing it from becoming electronically isolated or “dead,” which is a large source of capacity fade and low CE in lithium metal cells. This mechanism may be at work in these hybrid cells allowing them to achieve fairly good capacity retention and CE values even though the morphology is not optimal.

Hybrid Cells Tested Using Alternative Cycling Protocols

To this point, all of the hybrid cells described have been subjected to 100% state of charge (SOC) cycles to specified upper cutoff potentials. Under these conditions with mechanical pressure and an electrolyte optimized for lithium metal these cells can achieve approximately 80% capacity retention at 160 cycles. However, this graphite/lithium metal hybrid anode concept was based on the analogy to partial Si-graphite anodes. Cells utilizing these Si-Gr anodes demonstrate increased lifetime when cycled with less than 100% depth of discharge as this avoids fully accessing the silicon. Based on this concept we felt it would be useful to test the hybrid cells with protocols incorporating partial SOC cycles. For these hybrid cells, rather than limiting the depth of discharge, the upper cutoff voltage will be capped, that is, the cell would operate as a lithium-ion cell for the majority of its lifetime, while periodically accessing the lithium metal capacity with a “boosted” charge to 4.4 V.

Table 1 indicates that hybrid cells cycled to 100% SOC deliver an energy density of 890 Wh/L or about 20% higher than lithium-ion cells with the same positive

electrode. If an electric vehicle with a conventional lithium-ion battery can deliver a range of 400 km, then hybrid cells could enable a range of 480 km. By capping the upper cutoff voltage of hybrid cells to operate in lithium-ion mode, the average cell voltage and delivered capacity will decrease. As a result, operating a hybrid cell in lithium-ion mode delivers an energy density of 530 Wh/L, about 25% less than a conventional lithium-ion cell. This would result in a range of 300 km. In a study of driving behavior for EVs, Smart et al.³⁴ showed that only 1% of daily trips are longer than 325 km on average. Therefore, operating hybrid cells most of the time in lithium-ion mode enabling a range of 300 km, while periodically using the lithium metal portion for long > 400 km trips, as mimicked by this testing protocol, should be viable for most drivers.

This hybrid protocol uses an approximate midpoint of the voltages that were found to be the onset of lithium metal plating at 0 cycles and 50 cycles, since the plating voltage was not static, as discussed previously. Thus, the cells cycled to 4.0 V for 10 cycles (cycling very little, or no, lithium metal capacity depending on the cycle number) and charged to 4.4 V one time (indicated as 10 cyc/1 cyc in figure captions). This protocol was continuously repeated in both LDBF and 2F1L electrolyte cells. Additionally, the “boosted” protocol was also tested using 10 low voltage cycles/5 “boosted” cycles (indicated as 10 cyc/5 cyc in figure captions).

Figure 4A displays the hybrid cycling data with intermittent “boosted” cycles. After almost 400 cycles, the 10/1 protocol cells with applied pressure have lost very little of the original graphite capacity after charging to 4.4 V ~40 times. The plating and stripping of lithium metal multiple times on top of the graphite does not appear to be severely inhibiting the capacity of the graphite portion of the anode. In this way the cell can operate reasonably well in “lithium-ion mode” with periodic cycles accessing the lithium metal capacity. Figure S5B shows that this protocol is also enabled at 20°C after hot formation. Figure 4B shows the capacity retention of just the boosted charges to 4.4 V plotted with the capacity retention shown previously of hybrid cells on a standard 100% SOC protocol for comparison. The intermediate boosted charges track similarly to the slope of the standard cells with the LDBF electrolyte clearly outperforming the 2F1L formulation. These results suggest that the hybrid cell contains a certain number of 100% SOC cycles, which can be utilized periodically among many hundreds of cycles of the graphite anode.

A challenge for further improving hybrid cells will be developing electrolytes that are optimized for simultaneous graphite and lithium metal passivation. To ensure that the LDBF electrolyte used in this work to facilitate efficient lithium metal cycling does not negatively impact the graphite anode, we tested the performance of LDBF electrolyte in conventional lithium-ion cells. Figure S7 shows that an NMC532/graphite cell with LDBF loses 3% capacity after 500 cycles. This is compared to an optimized lithium-ion electrolyte with (1.2 M LiPF₆ EC:EMC (3:7) + 2% VC 1% ethylene sulfate (DTD), the electrolyte used in Harlow et al.³⁵), which loses 2% capacity under the same conditions. Therefore, we conclude that the LDBF is not negatively affecting the underlying graphite capacity in hybrid cells. An additional test was performed in which hybrid cells were cycled to 100% SOC 60 times between 3–4.4 V and then cycled between 3–3.95 V only accessing the graphite capacity in lithium-ion mode for a further 440 cycles (Figure S8). Negligible capacity loss was observed after 440 cycles in lithium-ion mode, again indicating that the cycling of lithium metal in hybrid cells does not have a negative impact on the graphite cycle life with LDBF electrolyte.

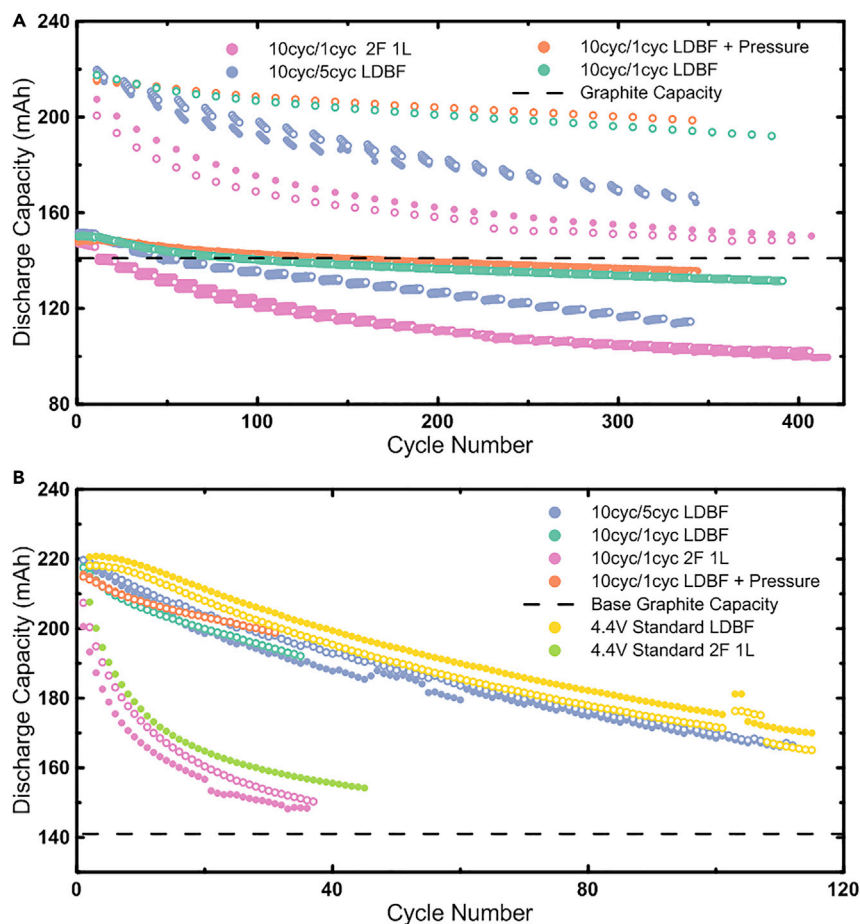


Figure 4. Performance Using a Realistic Mixed Lithium-Ion Mode/Hybrid Mode Cycling Protocol

(A) Capacity retention of cells cycled on a hybrid protocol charging to 4.0 V (graphite capacity) with periodic “boosted” full charge cycles to 4.4 V to access the lithium metal capacity; low voltage to high-voltage cycle ratio was either 10/1 or 10/5 as indicated.

(B) Capacity retention of just the “boosted” full charge cycles isolated from the full cycling data in the plot above, capacity retention of cells on the standard 100% SOC protocol are included for comparison. Open and closed symbols of the same color represent pair cells. (LDBF: 1 M LiDFOB 0.4 M LiBF₄ FEC:DEC 1:2; 2F1L: 1.5 M LiPF₆ EC:EMC:DMC 25:5:70 2 wt % FEC 1 wt % LFO).

Figure 4 shows the cells that charged with the 10/5 protocol quickly demonstrated the problem with using a static upper cutoff voltage for the graphite capacity. While the 4.4 V charges track similarly to the standard 4.4 V hybrid cells, the graphite capacity cycles appear to be rapidly decreasing with each discharge. This is due to the effect of the negative electrode slippage discussed previously. The voltage at which the graphite reaches its maximum capacity rises as the cells are cycled, and thus these cycles to 4.0 V fail to access the full capacity after a certain number of cycles. This observation also strengthens the theory that the slippage of the negative electrode is due to loss of lithium metal capacity, as the 10/1 cells do not exhibit this slippage as dramatically. In practice, the charging end condition for the graphite region may need to be based on capacity rather than voltage to have more precise control over the hybrid anode.

In addition to the intermittent cycling, cells were also cycled with a fast charge to the approximate limit of the graphite capacity, then cycled with the normal lithium metal

rate protocol in the approximate lithium metal region. The cycling data for these tests are shown in supporting Figure S9, and there is little penalty for cycling at a faster rate in the graphite region. In this way the cells become more flexible as the slow charge limitation of lithium metal is only imposed on a portion of the charging protocol.

Conclusions

In this work, we introduced the concept of a hybrid graphite/lithium metal anode, a novel anode system for lithium batteries, in which a graphite anode is reversibly cycled with lithium metal plating on top at high voltage to improve energy density. We found that these hybrid cells perform best when utilizing electrolytes optimized for lithium metal operation. When paired with conventional lithium-ion electrolytes the hybrid cells fall below 80% capacity retention in less than 15 cycles. Conversely, with the use of a LiDFOB/LiBF₄ dual-salt electrolyte and the application of uniaxial pressure, lithium can be reversibly cycled on top of the graphite with these cells achieving 160 cycles before falling below 80% capacity. With the LDBF electrolyte, lithium plated as large smooth grains on top of the graphite during formation, a morphology that was further enhanced with the application of pressure. This morphology deteriorated with cycling; however, a constant and fairly high CE for lithium metal cycling (99.6%) was maintained. For more practical applications of these batteries, we found that the hybrid cell can be operated primarily as a lithium-ion battery, cycling only to the limit of the graphite capacity, and charging to high voltage periodically when a higher capacity is required. This means that intermittent high energy cycles utilizing lithium metal can be dispersed among hundreds of conventional lithium-ion cycles where only the graphite is utilized. Operating in this way results in little penalty to the lithium metal capacity retention. This leads to the conclusion that the fade of lithium metal capacity is primarily dominated by the number of high-voltage cycles the cell has undertaken, and the number of cycles the graphite capacity has completed is of little relevance. These results demonstrate that with the right choice of electrolyte and cycling conditions, hybrid lithium-ion/lithium metal operation can be achieved simultaneously.

EXPERIMENTAL PROCEDURES

Resource Availability

Lead Contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the Lead Contact, Alex Louli (alex.louli@dal.ca).

Materials Availability

This study did not generate new unique materials.

Data and Code Availability

The datasets generated in this study are available from the Lead Contact on request.

Cell Preparation and Electrochemical Testing

Dry (no electrolyte) and sealed Li[Ni_{0.5}Mn_{0.3}Co_{0.2}]₂O₂ (NMC532)/graphite pouch cells were manufactured by Li-Fun Technology (Xinma Industry Zone, Golden Dragon Road, Tianyuan District, Zhuzhou City, Hunan Province, PRC, 412000). The NMC used in these pouch cells was single crystal NMC with a titanium-based coating.³⁶ The graphite used was Kaijin AML-400 artificial graphite from Kaijin, China.³⁷ The single side electrode loadings were 15.7 mg cm⁻² (46 μm) for the positive electrode and 5.0 mg cm⁻² (36 μm) for the negative electrode. These loadings and electrode thickness values, along with corresponding comparisons to

conventional Li-ion cells and anode-free cells, are presented in Table S1. The positive and negative electrodes consisted of 94% and 95.4% active material, respectively. Cells were transferred into an argon-filled glove box, cut open, and dried at 100°C for 14 h under vacuum and returned to the argon environment. Cells were then filled with 0.85 mL of their respective electrolyte, resealed, and removed from the glove box. Electrolyte solutions were prepared from pre-mixed solvents with the addition of salts and additives in an argon-filled glove box. The cells were held at 1.5 V for 24 h to assure complete wetting of the porous separator, and were then charged at C/20 to the upper cutoff voltage before returning to 3.8 V as formation step at 40°C.

Cells were cycled using various charging rates, upper cutoff voltages, electrolytes, and other changes to cycling protocol. The cells were cycled with a C/5 charge rate and C/2 discharge rate unless otherwise specified. The lower voltage cutoff in all cases was 3.0 V while various upper voltage cutoffs were utilized including 4.2, 4.3, or 4.4 V. Cycling was conducted in temperature boxes held at 40°C. Cells that were testing slower charge rates charged at C/10 with a constant current. The majority of the cells were cycled in a plastic enclosure that maintained a low pressure of approximately 75 kPa. Certain cells were cycled under the high-pressure condition in a rigid aluminum enclosure with the application of 1,200 kPa uniaxial pressure.

Cells were also tested with a hybrid protocol in which they were charged to 4.05 V accessing only the graphite capacity with periodic 100% SOC cycles to 4.4 V that also accessed the lithium metal capacity. The ratio of the low voltage to full SOC cycles was 10/1 or 10/5 as indicated.

SEM

Cells after formation and after 20 cycles, with or without pressure applied, were disassembled in an argon-filled glove box after being removed from cycling. The anodes were cut into sizes that were appropriate for the SEM sample holder and rinsed with DMC to remove any debris. The samples were then dried and attached to SEM sample stubs via carbon tape and transported in an argon-filled container to the SEM chamber for imaging. A Hitachi S-4700 field emission scanning electron microscope was used for this work. The SEM was used on secondary electron imaging mode with an electron accelerating voltage of 3.0 kV.

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.joule.2020.04.003>.

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AUTHOR CONTRIBUTIONS

J.R.D., M.G., A.J.L., C.M., and R.W. conceived of the idea. C.M. carried out the measurements and analysis. M.G. and A.J.L. wrote the manuscript. J.R.D. edited the manuscript and interacted with the other authors regarding experiment design and data analysis.

DECLARATION OF INTERESTS

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